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A survey of cloud computing data integrity schemes: Design challenges, taxonomy and future trends

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ABSTRACT

Cloud computing has gained tremendous popularity in recent years. By outsourcing computation and storage requirements to public providers and paying for the services used, customers can relish upon the advantages of this new paradigm. Cloud computing provides with a comparably lower-cost, scalable, a location-independent platform for managing clients' data. Compared to a traditional model of computing, which uses dedicated in-house infrastructure, cloud computing provides unprecedented benefits regarding cost and reliability. Cloud storage is a new cost-effective paradigm that aims at providing high availability, reliability, massive scalability and data sharing. However, outsourcing data to a cloud service provider introduces new challenges from the perspectives of data correctness and security. Over the years, many data integrity schemes have been proposed for protecting outsourced data. This paper aims to enhance the understanding of security issues associated with cloud storage and highlights the importance of data integrity schemes for outsourced data. In this paper, we have presented a taxonomy of existing data integrity schemes use for cloud storage. A comparative analysis of existing schemes is also provided along with a detailed discussion on possible security attacks and their mitigations. Additionally, we have discussed design challenges such as computational efficiency, storage efficiency, communication efficiency, and reduced I/O in these schemes. Furthermore, we have highlighted future trends and open issues, for future research in cloud storage security.

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1. Introduction

Cloud computing provides a flexible and cost-effective solution for many services through Internet (Nepal et al., 2011). It is considered as a major Information Technology (IT) shift and

a latest model of computing over pooled computing resources such as bandwidth, storage, servers, processing power, services, and applications. Today, this new model has gained tremendous popularity and is receiving a lot of attention from researchers in the academic and industrial communities. Essential characteristics of the cloud-computing model include

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on-demand self-service, rapid elasticity, resource pooling and broad network access. Cloud computing has gained a lot of popularity, which is mainly due to the following reasons, as discussed in [Armbrust et al. \(2009\)](#): (a) Cloud computing has eliminated the overhead of planning from the user, providing resources that are available on-demand, self-service, and the ability to scale according to requirements. (b) Cloud computing has eliminated up-front commitment by the end users. Pay-as-you-go model has allowed companies to start small and increase their computing resources only when needed. The deployment models used by cloud computing are Private Cloud (internal datacenters of business or organizations which are not available for general public use) ([Armbrust et al., 2009](#)), Public Cloud (private cloud made accessible for general public use on pay-as-you-go model) ([Armbrust et al., 2009](#)), and Hybrid/Multi-Cloud (cloud computing environment in which an organization provides and manages some resources in-house, and the other services are provided externally) ([Zhu et al., 2011](#)). Cloud computing services are organized as Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS). Under the umbrella of these services a number of new services have also emerged, e.g. Data Integrity as a Service (DIaaS) ([Nepal et al., 2011](#)), Database as a Service ([ALzain and Pardede, 2011](#); [Arasu et al., 2013](#); [Curino et al., 2011](#); [Seibold and Kemper, 2012](#)), Logging as a Service ([Frenot and Ponge, 2012](#); [Zawoad et al., 2013, 2015](#)), Provenance as a Service ([Hammad](#)

and [Wu, 2014](#)), Security as a Service ([Al-Aqrabi et al., 2012](#); [Hussain and Abdulsalam, 2011](#)), Big Data as a Service ([Zheng et al., 2013](#)) and Storage as a Service ([Calder et al., 2011](#); [Nabeel and Bertino, 2012](#)). An overview of cloud architecture is shown in [Fig. 1](#).

Cloud Storage as a service is a growing trend with features like elasticity, pay-as-you-go, business continuity with long-term retention and risk mitigation through disaster recovery ([Nepal et al., 2011](#)). Not all these features are available with on-premises storage. Popular cloud-based storage services available today are Dropbox, One Drive, Amazon S3, Google Drive, Box, and Sugar Sync etc. Nowadays, to improve business strategies, organizations use analysis techniques over their historical data ([Liu et al., 2014](#)). Some business sectors, for instance telecom and e-health, have compliance requirements ([Luxton et al., 2012](#)), which bind them to keep historical data over a specified period. Not every organization is equipped to manage large secondary storage or build their private data centers (because of the cost associated with building and maintaining such infrastructure). Cloud storage can be of great service to such organizations because of its flexible model. However, the loss of control is an inherent issue with outsourced data storage model. Although the cloud service provider (CSP) is bounded by a service level agreement (SLA) to ensure data security, users cannot solely rely on such agreements. Furthermore, reliance on a contractual obligation may fail to detect

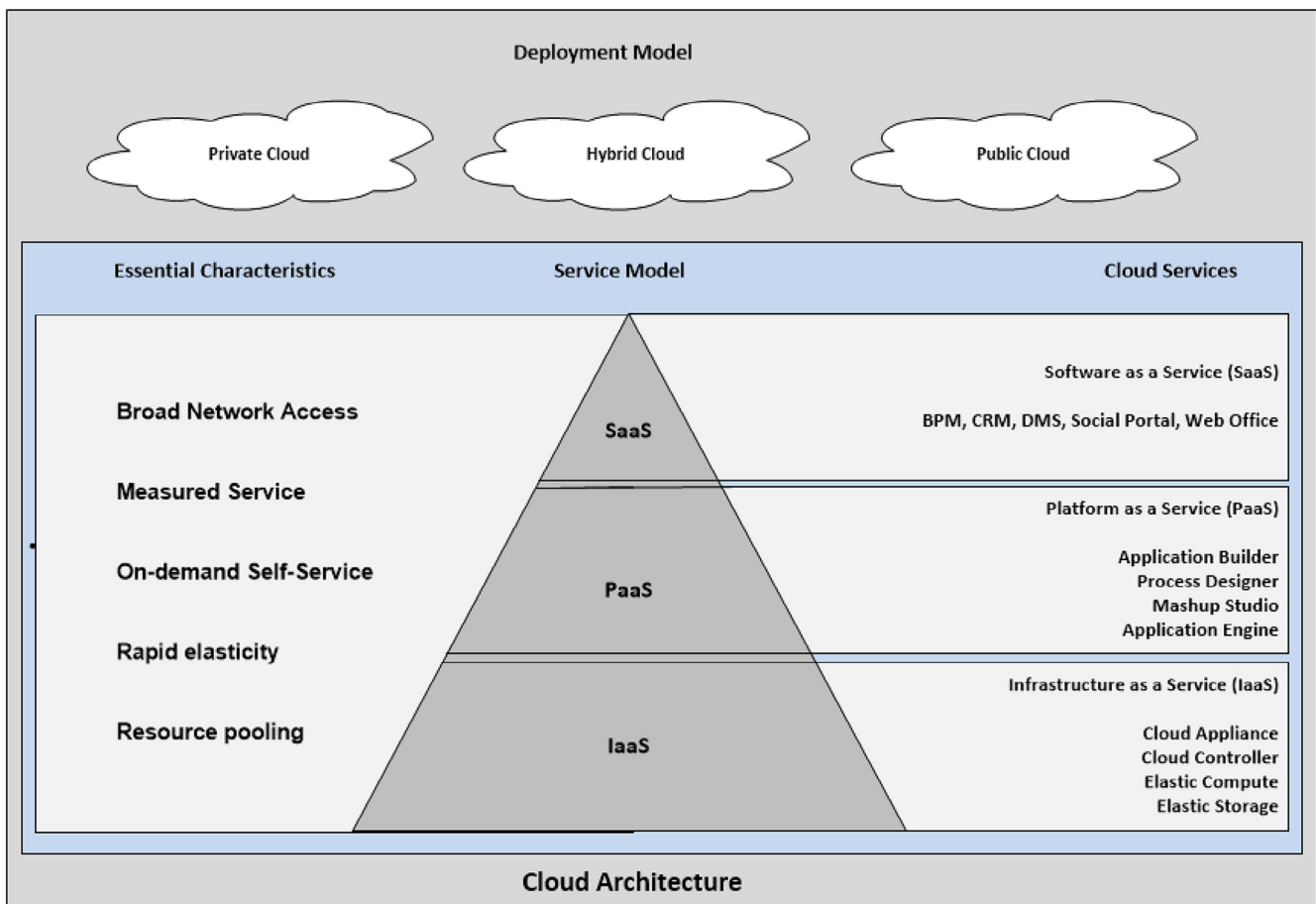


Fig. 1 – Overview of cloud architecture.

the malicious behavior of the service provider. Cloud computing operational details are not transparent to the customers and the CSP may be untrusted. Therefore, besides the convenience provided by cloud model, data security issues such as confidentiality, privacy, and data integrity are also associated with cloud storage service model. Data can be manipulated or lost due to accidental or intentional malicious activity, which can be a nightmare for the user and an embarrassment for cloud service provider. Cloud has a provision of “multi-tenancy” i.e. cloud resources will be shared and utilized by multiple users; therefore, adversaries can take advantage of vulnerabilities in the cloud (Ristenpart et al., 2009). Administration errors can also damage data; for example, failures in data migration or backup/restore process. Consequently, data integrity is at the core of outsourcing data over the cloud storage. Many data integrity schemes have been developed for ensuring correctness of outsourced data in recent years. When the integrity of outsourced data is to be checked, it is not feasible to download all data from the remote server and verify as it incurs high communication and computational cost. To avoid this high cost, most of the data integrity schemes perform *blockless verification* (Ateniese et al., 2007). This is done without downloading actual data from the cloud, rather it is based on some metadata, which is generated by using original data before it is outsourced. Later on, using this metadata, proofs for integrity verification are generated by cloud service provider and verified by the data owner. Research on data integrity schemes started from work on static data or append-only data and then extended to dynamic data (supporting CRUD operations on block level) (Ateniese et al., 2008). Initially, private (single user/data owner) verifiability was supported and later on, public (multiple users) verifiability (Wang et al., 2012) was introduced with or without third party auditor (TPA). Privacy issues such as leakage of data and user anonymity were identified due to the introduction of TPA. Thus, privacy preserving data integrity schemes (Erway et al., 2009; Wang et al., 2012) were developed to overcome the privacy concerns of the users. Application of these techniques extended from a single copy to multiple copies of data (Krzywiecki and Kutty, 2012), constraint-based data geolocation (CBDG) (Rashid et al., 2013), multi-cloud (cloud of clouds) (Cash et al., 2013), data deduplication (Zheng and Xu, 2012) and proof of ownership (Zhu et al., 2012). Researchers used homomorphic tags, bilinear pairing (Wang and Zhang, 2014), algebraic signatures (Yu et al., 2014), fountain codes (Shi et al., 2013), erasure codes (Chen, 2013), RS codes based on Cauchy matrices (Chen and Curtmola, 2012b) and other techniques as metadata for integrity verification. Efficiency being a real challenge in data integrity schemes is measured regarding computation, communication and storage costs incurred. One other desirable characteristic of data integrity is unbounded queries (which mean no restriction on how frequently user can verify data). When data size is large, this desirable characteristic is hard to achieve (Ateniese et al., 2008). Achieving efficiency sometimes results in a compromise on security (Chen, 2013). Implementation primitives also impact computation, communication and storage efficiency. Asymmetric cryptographic primitives provide strong security, but efficiency decreases due to complex computation and large keys. This decline in efficiency becomes more prominent for large datasets. Disk I/O is another crucial factor

affecting the efficiency of the verification process in data integrity schemes. For allowing data sharing over cloud storage, strong data access control policies are needed to ensure data security. The requirements of a data integrity scheme vary according to the deployment environment. Schemes designed for a single copy of data are not applicable within server and cross server replication (Curtmola et al., 2008). The requirements of data integrity scheme change with cloud deployment model, e.g. hybrid or multi-cloud (Zhu et al., 2011), whereby the scope of data integrity is extended. Moreover, Service Level Agreement (SLA) checking is becoming a fundamental part of data integrity.

Existing surveys (Oualha et al., 2012; Sookhak et al., 2014; Worku et al., 2012) on data integrity schemes have performed only a comparative analysis based on their characteristics and discussed their internal implementation. These studies lack general guidelines about (i) the key parameters which control overall efficiency of data integrity schemes and (ii) designing of an efficient, scalable and secure data integrity scheme.

The key contributions of this paper are:

1. We define a taxonomy, based on the attributes of data integrity schemes; for instance, verification approach, nature of data they deal, nature of metadata they use, implementation primitives used and deployment scenarios, to describe all aspects of data integrity.
2. We identify design challenges of data integrity schemes and highlighted a set of parameters, which control their overall efficiency. We analyze the impact of these parameters, which can be helpful in designing scalable and efficient schemes.
3. We summarize the security aspects (possible attacks due to vulnerabilities) of data integrity schemes as well as mitigation techniques.
4. We identify the future trends (e.g. geolocation, assured deletion, computation assurance, deduplication) of research in the context of data integrity schemes.
5. We perform a comparative analysis of the most recent existing data integrity schemes based on identified characteristics.

1.1. Organization of the paper

The paper is organized as follows. In Section 2, cloud storage issues are discussed. Section 3 presents a thematic taxonomy of data integrity schemes. In Section 4, common characteristics are discussed and a comparative analysis of data integrity schemes is provided. Section 5 covers the design challenges of data integrity schemes. Future research directions in data integrity schemes are presented in Section 6. Section 7 concludes the paper.

2. Cloud storage issues and need of data integrity schemes

Cloud storage is beneficial for both users and service providers. Cloud users incur minimum capital investment and are relieved from infrastructure maintenance responsibility (Nepal

et al., 2011). By utilizing cloud storage, a user is provided location-independent access to services (Armbrust et al., 2009). Meanwhile, a CSP gains high revenue over available infrastructure by sharing it among multiple users. This high level of resource utilization leads to optimized energy consumption. Despite these benefits, cloud storage has some inherent weaknesses, as CSP is always assumed untrusting, which may have malicious motivations. A CSP may behave maliciously either intentionally (charging for all data and deleting the unaccessed data after usage statistics analysis or keeping the fewer replicas as per commitment) or unintentionally (service provider himself is unaware if bad sectors created in disk or hard disk crashes). The user's concerns about cloud storage are given below (Chow et al., 2009; Sookhak et al., 2014; Wei et al., 2014):

1. Unauthorized physical access

In cloud-based storage, data may be scattered across multiple servers and locations. The user loses control over his data and is unable to inspect the data links visually. Therefore, the risk of unauthorized physical access is one of the issues of cloud storage. Re-allocation of storage space, disposal of old equipment or reuse of drives can result in data breaches.

2. Cloud storage provider espionage

Users are usually concerned that service providers may attempt to obtain sensitive information from their data without his consent or knowledge. Especially when data contains confidential information such as credit card numbers, medical data, etc., users hesitate to adopt cloud storage.

3. Data lock-in

No standards exist that define how data should be stored and maintained in the cloud. Every CSP has its proprietary format for data storage. Therefore, users get bound to a particular CSP and switching to another is not simple. Frequent changes in cloud-based services also cause problems for users.

4. Cloud storage provider stability

Cloud storage claims to provide high availability. However, what about the CSP stability? This includes:

- How many services provided by a CSP are stable and how frequently do they change?
- Is the CSP capable of mitigating natural disasters?
- Reliability of other company's services hired by the CSP?
- What if the CSP goes bankrupt?

All these questions can result in user hesitation for the adoption of cloud storage.

5. Accessibility effectiveness or high availability

The effectiveness of data access from the cloud depends on the performance of network and WAN bandwidth. The choice of network hardware and algorithms used by the CSP and the

level of precautions taken affect the reliability and availability of data. One major obstacle in the adoption of public cloud from the user perspective is high availability, as pointed by authors in Armbrust et al. (2009). The assurance of high availability and QoS is part of SLA, and the end user does not have to worry about how it is provided. For example, CSP may have agreed with the user to provide a level of 99.95% availability, i.e. 4.38 hours of downtime is to be expected in a year.

6. Security against internal and external network attacks

When user data leaves the premises, the risk of attack it is subjected to is also increased. At the CSP end, user's data is insecure and exposed to internal and external attacks during data transmission and retrieval. Malicious users can damage other user's data as cloud resources are shared among multiple users. Faulty equipment, hardware or software bug or criminal activity may result in data security breaches. Secure channels are used to ensure that the data transmission is protected. Furthermore, encryption of data also helps to preserve data confidentiality.

7. Cloud computation reliability

How can a user be ensured that CSP is faithfully performing the computations and giving correct results? Trusted hardware or software is used which provides proof of computation authenticity.

8. Cloud forensic investigation: a real challenge

Digital forensics investigation is a challenging task for any security incident due to the multi-tenancy environment and decentralized data processing in cloud computing. Traditional techniques of investigation and evidence collection are not practical in the context of a cloud environment. National Institute of Standards and Technology (NIST) defines "Digital Forensic" as a science of identification of an incident, collection, examination and analysis of evidence data while keeping the integrity of evidence information intact and maintaining the strict chain of custody of data (Kent et al., 2006). The nature of forensic evidence data varies according to cloud services and deployment model. From a technical perspective, evidence data can be at "rest", "motion" or "execution" state (Birk, 2011). Usually, data on disk storage are considered to be in the rest state. Virtual machines' activity logs are an example of evidence data at rest and could be used for digital forensics (Zawoad et al., 2015). Data traveling over the network from one entity/device to other is considered to be in motion. Irrespective of the cloud deployment model, the following three components serve as a potential source for evidence data in cloud forensic (Birk, 2011):

- a. **Virtual instance:** A virtual instance is assigned to each user in the cloud and most of the security incidents happen on this virtual instance. Therefore, it holds evidence data of any security breach and could be analyzed by both the user and CSP. The forensic investigation becomes a real challenge in SaaS and PaaS cloud deployment models, as access to the virtual instance for evidence data collection is either very limited or none at all.

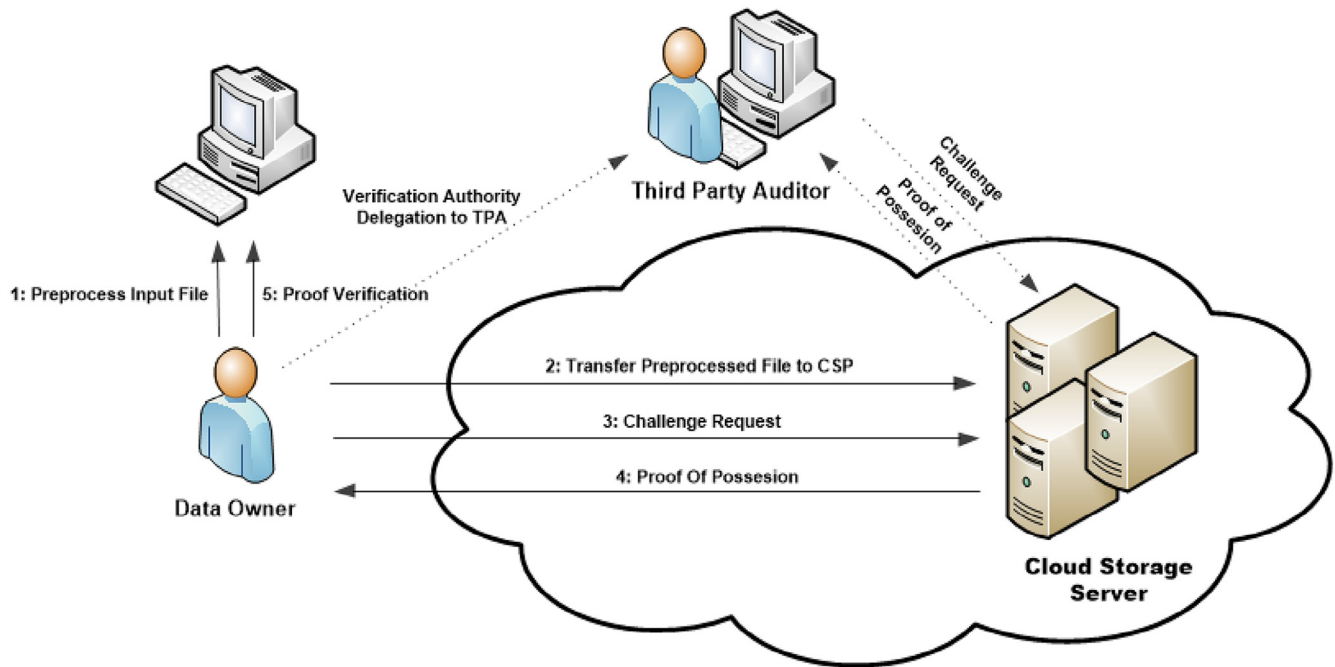


Fig. 2 – Data integrity scheme protocol.

- b. **Network layer:** Network component logs contain evidence data for any security breach. Different OSI/ISO network layers facilitate by providing information on protocols and communication for instances within the cloud as well as outside the cloud.
- c. **Client system:** The location of potential forensic evidence data on a client's system layer entirely depends on the selected cloud deployment model (IaaS, PaaS, SaaS). For example, a client's browser is an application that is communicating with SaaS services hosted in the cloud. For any security incident, forensic data collected from the browser environment of the client is valuable for exhaustive forensic investigation. However, the integrity of forensic data itself is a challenge.

Despite all the issues mentioned above, the user's primary concern is that security; confidentiality, privacy, and data integrity are not compromised. The CSP is bounded by SLA and standards such as HIPAA (Luxton et al., 2012), which are available for user's data security. Nevertheless, there is a need for mechanisms to ensure data integrity, which is considered as the essential requirement for cloud storage adoption.

3. Data integrity schemes

Data integrity schemes, discussed in this survey, are of probabilistic nature, tamper evident (not tamper resistant) and, are applicable only to secondary storage (non-volatile data). Integrity schemes for volatile data are out of the scope of this survey. There are three entities in a data integrity scheme: (i) Data owner who outsources his data, (ii) a cloud storage provider (CSP) to whom data are outsourced and (iii) an auditor who verifies the integrity of the data. An auditor can be the

data owner himself or he can delegate this responsibility to Third Party Auditor (TPA). Fig. 2 shows the working of a data integrity scheme protocol. There are two phases in a data integrity protocol:

1. **Preprocessing phase:** data are preprocessed, and some additional metadata are generated. Then, both the data and metadata are outsourced to the CSP.
2. **Verification phase:** an auditor sends a challenge request to the CSP who generates the proof of possession using the data and metadata and provides it to the auditor. Then, the auditor verifies the proof, to ensure that the data integrity is intact.

With the help of data integrity schemes, any data corruption or deletion can be timely identified and thus necessary measures can be taken to recover the data. To develop a better understanding of data integrity schemes, we have defined a thematic taxonomy based on the nature of data they deal, the nature of metadata they use, their capabilities, implementation primitives used and deployment scenarios.

3.1. Taxonomy of data integrity schemes

The following are the attributes taken into consideration for defining the taxonomy of data integrity schemes. These attributes not only describe the capabilities of any scheme but also give an idea about their limitations in practical scenarios. Fig. 3 shows the summarized view of schemes with reference to these attributes.

- **Approach:** The approach refers to the nature of guarantee provided by a data integrity scheme, which can be deterministic or probabilistic. Deterministic schemes (Deswarte

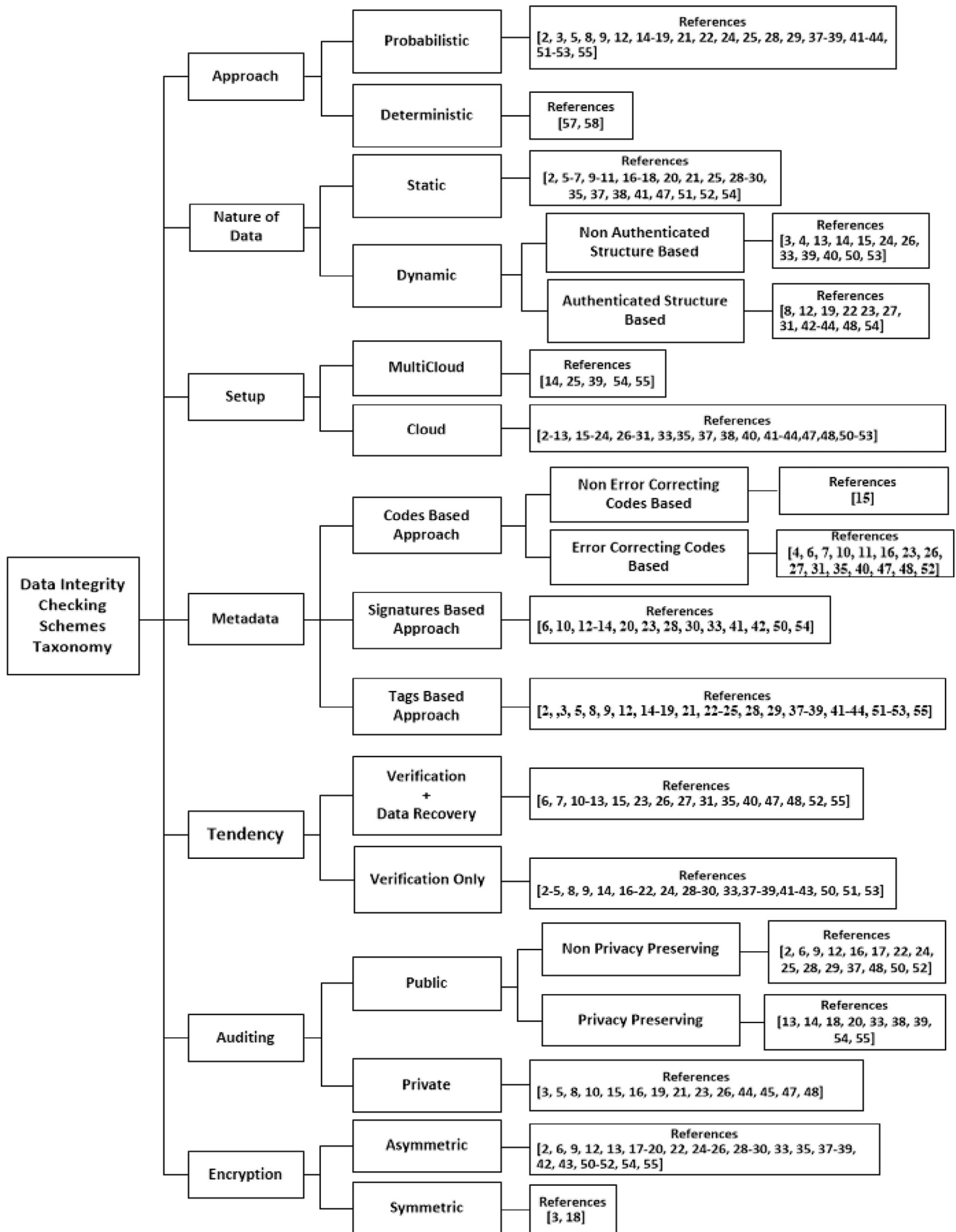


Fig. 3 – Taxonomy of data integrity schemes.

et al., 2004; Han et al., 2014) need to access the complete file to determine the integrity and give 100% possession guarantee. Whereas, probabilistic schemes (Ateniese et al., 2007, 2008) use randomly chosen blocks of data to verify the integrity and give less than 100% guarantee. Deterministic schemes are not suitable for large size files, e.g. archives having data size in GBs or TBs as integrity verification of such a file takes too much time. This can lead to a limitation on how frequently the user can perform integrity verification. Therefore, probabilistic schemes are more appropriate for large files. On the other hand, deterministic verification is suitable for smaller sized files having the size in few megabytes. Digital libraries (e-books, articles, research papers) and media contents (songs, pictures, etc.) normally have small size files but are large in number. For such type of data, deterministic schemes are ideal to provide 100% possession guarantees. Ateniese et al. (2007) proposed the first probabilistic integrity scheme in 2007, which used random sampling of blocks instead of reading the whole file. By inspecting 4.6% of total blocks of a file, it provides 99% possession guarantee. This scheme provided the base for probabilistic integrity checking schemes, but its performance decreases as the size of data increase.

- **Nature of data:** This attribute describes the nature of data, on which the scheme is applicable. Data can be of static nature (Yuan and Yu, 2013; archival data, backups, or data that are never modified but are appended only) or of dynamic nature (Curtmola et al., 2008; Zhang and Blanton, 2013; that frequently change due to operations like create, read, update and delete). The integrity of volatile data is not considered in this survey. The nature of data also plays an import role in scheme evaluation. The scheme proposed in Ateniese et al. (2007) is only suitable for static data, because it requires re-computation of complete file tags when new data are inserted in between existing data. Similarly, the scheme (Ateniese et al., 2008) provides support for modification, deletion and append operations; however insertion operation is not supported. The scheme in Erway et al. (2009) provides support for dynamic data operations such as insertion, updating, deletion, and modification. Furthermore, the scheme (Erway et al., 2009) also kept the fundamental properties of verification (unforgeability, unbounded queries) intact. The use of authenticated skip list based FlexList to support dynamic operations was proposed in Esiner et al. (2013). Dodis et al. (2009) improved the efficiency of POR schemes for static data. Bowers et al. (2009b) proposed a POR scheme for static data having low storage overhead and providing higher tolerance rate for errors. The highlighted feature of their proposed design was an incremental encoding of data blocks enabling the support for handling large size datasets. Barsoum and Hasan (2012) extended the integrity checking of static data for multiple copies. The strength of their proposed scheme is that verification time is independent of the number of copies. Stefanov et al. (2012) introduced a framework named “Iris” giving cloud users high assurance of data integrity, data freshness and data recovery in case of any corruption.
- **Setup:** The setup refers to the nature of environment in which data integrity scheme will be deployed. Since data are placed over the cloud, which can be simple (self-sufficient,

providing all types of capabilities itself) or “hybrid/cloud of cloud/multi-cloud” (managing some resources internally and other are provided externally; Zhu et al., 2011), nature of setup leads to some additional requirements in data integrity schemes. For example, a cloud setup can support replication within a server and across servers, extending the integrity checking from single to multiple replicas of files. Curtmola et al.’s (2008) proposed scheme is called “multiple replicas provable data possession scheme (MR-PDP)”, which is capable of generating new replicas on demand in case of any corruption or deletion. Application of MR-PDP is limited to a single cloud having multiple storage servers but is not extendable to a multi-cloud environment. Bowers et al. (2009a) focused on high availability by exploiting the redundancy within a server and across servers. Peterson et al. (2011) extended data integrity and added support for verification of SLA to ensure that outsourced data are in specified geographical boundary. Zhu et al. (2012) introduced characteristics of the high security, high performance, and transparent verification for PDP model in the context of multi-cloud.

- **Tendency:** The tendency of a data integrity scheme can be verification only (identification of corruption/deletion of data) or verification with data recovery (if any corruption is identified). From the perspective of tendency, data integrity schemes are categorized into provable data possession (PDP) or proof of retrievability (POR) (Xu and Chang, 2012).
 - a. **Provable Data Possession (PDP):** PDP schemes are probabilistic as they use the sampling of random blocks instead of reading the whole file for verification. In PDP, the original data are preprocessed to generate some metadata. These metadata are placed with original data and is used later to verify the integrity of a user’s data. These schemes can only identify corruption in data but do not support-corrupted data recovery.
 - b. **Proof of Retrievability (POR):** POR is very much similar to PDP, but it also provides data recovery. POR schemes use the redundant encoding of data and hence provide recovery in case of failure. In other words, a PDP scheme can be transformed into POR by using error correcting or erasure codes. The auditing protocol of POR provides the guarantee that CSP is holding all the data of client and is still retrievable. However, PDP only ensures that the server is holding most of the data of a particular client, due to the probabilistic nature of the algorithm used in verification.
- **Metadata:** All data integrity schemes use some additional metadata along with the original data, which are utilized in integrity verification process. These metadata can be “tags” (Yuan and Yu, 2013; Zhang and Blanton, 2013; Zhu et al., 2012), having homomorphic verifiable property, which help in generating an aggregated value in the verification process. The metadata may be “signatures” (Luo and Bai, 2011; Tate et al., 2013), which are used as an alternative to tags, e.g. algebraic signatures to improve techniques like Reed–Solomon codes. Finally, metadata may consist of network codes that also support data recovery instead of tags and signatures, which may be used for identifying small (bit or byte level) data corruptions. For example, RSA-based homomorphic tags are used as verification metadata in

Ateniese et al. (2007), “HAIL” (Bowers et al., 2009a) scheme provides data recovery capability using integrity-protected error-correcting code (IP-ECC). Chen et al. (2010) utilized network codes to provide reduced storage overhead in distributed storage system used for backup and archival data. Okamoto protocol based signatures have been used in the scheme (Mohan and Katti, 2012) as metadata. Ateniese et al. (2011) improve the capability of PDP scheme by introducing robustness using Forward Error-Correcting Codes (FEC).

- **Encryption:** Over the year’s data, integrity schemes have utilized both symmetric (Ateniese et al., 2008; Lee and Chang, 2011) and asymmetric encryption (Barsoum and Hasan, 2012; Chen et al., 2013; Luo and Bai, 2011; Shen and Tzeng, 2011; Tate et al., 2013; Wang, 2013; Yuan and Yu, 2013; Zhu et al., 2011, 2012) for security. Ateniese et al. (2008) achieved scalability and efficiency by using symmetric key encryption and cryptographic hash functions. Ateniese et al. (2007) used RSA based homomorphic verifiable tags (HVT) as the metadata. HVTs are unforgeable and are used in blockless verification by which a server can construct a proof of possession and the client can verify it to ensure data integrity. Curtmola et al. (2008) utilized homomorphic tags of Ateniese et al. (2007) for multiple replicas and used a Pseudo Random Function (PRF) for masking. Wang et al.’s (2009) proposed a scheme for proof of retrievability using BLS signatures supporting public verifiability along with dynamic data operations. Zhu et al.’s (2012) proposed scheme is based on homomorphic verifiable response (HVR) and hash index hierarchy (HIH) and presents a cooperative PDP (CPDP) scheme from bilinear pairings to provide knowledge soundness. Lee and Chang (2011) proposed a hybrid scheme using both symmetric and asymmetric encryption for efficient data integrity verification protocol. Wang’s (2012) scheme is based on group signatures and Homomorphic MACs, providing integrity verification of data shared among large user groups.
- **Auditing:** Data integrity schemes either allow public (conducted by a third party auditor or multiple users) auditing or private auditing (performed by data owner). In the case of public auditing, privacy issues such as data leakage, user anonymity, etc. are associated. Public auditing schemes are further categorized into privacy preserving and non-privacy preserving schemes. Ateniese et al. (2009) proposed a public verifiable scheme, which is called Proof of Storage (POS). However, this scheme was vulnerable to privacy attacks. The same author proposed another POS scheme (Lee and Chang, 2011), which provided privacy protection. Krzywiecki and Kutty’s (2012) scheme utilized secure pseudorandom numbers (SPRN) with Lagrangian interpolation and provides broadcast encryption with public auditability for multiple users. Shacham and Waters (2008) have used BLS signatures to provide public auditing. Public auditing with privacy protection is achieved along with traceability, which means that the original user can trace a signature on a block and reveal the identity of the signer. Another significant contribution of this work is that it supports a large number of users without affecting the performance of the verification process (Wang et al., 2012). Shen and Tzeng’s (2011) scheme supports delegation of auditing authority; however, there is a restriction of re-delegation. Therefore, a user authorized for auditing cannot re-delegate the

authority to others. Wang (2013) has discussed the issues of public auditing and proposed provable secure and efficient proxy provable data possession (PPDP) scheme for cloud storage.

In this thematic taxonomy, we have described the attributes over which data integrity schemes can be categorized. The details of scheme’s workings have not been discussed here. See section 4.1 for comparative analysis of schemes.

4. Characteristics of data integrity schemes

All data integrity schemes possess a subset of following properties:

- **Remote verification:** Remote verification means that a verifier should not retrieve file blocks from the remote server for verification. This property is also termed as blockless verification (Ateniese et al., 2007). Blockless verification is an essential requirement for all data integrity schemes, as retrieving complete data file (especially with large sized files) on a client machine and verifying integrity is not practical. In blockless verification, the cloud accesses block of a data file and generate a proof of possession, which is later verified by a client to verify data integrity.
- **Unrestricted challenge frequency:** Unrestricted challenge frequency means that there should be no restrictions on the numbers of challenges made by a client or auditor to verify the integrity of remote data (Ateniese et al., 2007). Unrestricted challenge frequency property is also called unbounded queries. Integrity verification of outsourced cloud storage data is not a one-time activity. A client executes the verification protocol with intervals to timely identify any data corruption. The frequency of challenge requests is directly affected by the computation efficiency of the data integrity scheme. If verification process of a data integrity scheme is computationally expensive, then the client will use it less frequently and as a result unbounded challenge frequency would suffer.
- **Soundness:** An untrusted server should not be able to deceive (pass auditing without holding data) a challenge request (Wang and Zhang, 2014; Zhu et al., 2012). The soundness property of data integrity schemes ensures data reliability. The nature of a data integrity schemes is tamper evident, not tamper resistant. Therefore, if metadata are tampered with, or data get corrupted intentionally by the CSP or unintentionally, this should be timely identified by a data integrity scheme. If the CSP can pass a challenge request without holding the data or with corrupted data, then a client will never be able to identify data corruption promptly, and the value of the data will be lost. Therefore, the reliability of data directly demands the soundness of a data integrity scheme for cloud storage.
- **Stateless verification:** Neither a client nor the server needs to store the results of previous audits to verify future audits. Every challenge request is independent of all past verifications performed by the client. Stateless verification is indirectly an essential requirement of data integrity schemes. Otherwise, it will lead to a situation where state keeping

itself would become a burden either for client or for CSP. Keeping states of previous audits is unnecessary and might incur some cost regarding resources, e.g. memory or disk storage.

- **Robustness:** Robustness is the ability to identify even the smallest size of data corruptions (Ateniese et al., 2011; Chen and Curtmola, 2012a, 2012b). Data integrity schemes of deterministic nature have a limitation of computational efficiency and are not applicable for large datasets. To overcome these limitations, a probabilistic approach is adopted. Probabilistic data integrity schemes work on randomly chosen samples of data for integrity verification, which raises the issue of identification of small (bit or byte level) corruptions. Robustness property enhances the soundness of data integrity schemes by providing the identification of minor level corruptions in data.
- **Data recovery:** For a data integrity scheme, it is not sufficient just to identify the misbehavior by the remote storage server. As cloud users are also interested in, complete data recovery to avoid an unpleasant situation. Data integrity schemes are divided into two categories depending on their tendency. All data integrity schemes provide identification of corruptions (Ateniese et al., 2007; Yuan and Yu, 2013), but some also support data recovery along with identification (Bowers et al., 2009b). Usually, error-correcting codes (ECC) are used by the schemes providing data recovery (Bowers et al., 2009a; Chen and Curtmola, 2012b). The network codes have been utilized by (Chen et al., 2010) to provide better efficiency in comparison with erasure codes for archival systems.
- **Dynamic data handling:** Data can be either static (backup or archival data) or dynamic nature (supporting operations like insertion, deletion, and modification). Providing data integrity for dynamic data is more challenging than for static data or append-only data. Most of the schemes proposed in the literature are not capable to handle dynamic data. Dynamic data handling characteristic (Curtmola et al., 2008; Zhang and Blanton, 2013) demands that the data integrity should remain intact, even after insertion, deletion or modification operation is performed.
- **Public/Private auditability:** There are two approaches to audit the data stored at the remote server. The first method allows the data owner to do the audit. The second method allows a third party auditor to audit. The audit process in both approaches is performed without retrieving the remote data (Ateniese et al., 2009; Luo and Bai, 2011). In both approaches, the audit is performed without retrieving the remote data (Ateniese et al., 2009; Luo and Bai, 2011). If only the data owner can verify the integrity of the outsourced data, then this scheme is considered to provide private auditability. However, in some cases, it is not practically feasible for the data owner to remain online all time for data integrity verification. Hence, the data owner can delegate this responsibility of integrity verification to a third party auditor or other users. A data integrity scheme must support public auditability property to support this audit delegation.
- **Privacy preserving:** During the audit, the verification process should protect data privacy. Privacy (Wang et al., 2010; Zhu et al., 2010) concerns are introduced due to public verifiability. As the data owner will not allow the disclosure of

his private data to a third-party auditor, the privacy preservation property demands that a third party auditor should not obtain any confidential information about the user's data but can still verify the integrity of outsourced data.

- **Fairness:** Fairness means that a data integrity scheme should provide protection to an honest CSP against legitimate but dishonest users, who may attempt to accuse a CSP of manipulation in the outsourced data (Zhu et al., 2011). If a data integrity scheme does not support fairness, it means dishonest users can damage CSP reputation.

Based on the aforesaid common characteristics, we conduct a comparative analysis of data integrity schemes in subsection 4.1.

4.1. Comparative analysis of data integrity schemes

In this section, we will provide a comparative analysis and comparison of data integrity schemes, as illustrated in Table 1.

Ateniese et al.'s (2007) proposed technique have used the RSA-based homomorphic tags as verification metadata. The said tags enabled blockless verification and are unforgeable. However, the scheme is applicable only to static data or appends only data. In the case of dynamic data operations, if any block is inserted or deleted from within the file, tags re-computation is required for all blocks ahead of modification point, which is not practical. PDP scheme proposed public verifiability with weaker possession guarantees and posed no limitation on the frequency of challenges. In 2008, these limitations of dynamic data operations in PDP were partially handled in Ateniese et al. (2008), by allowing modification, deletion and append operations. Support for insertion operation was still not supported by this technique. Scalability and efficiency were achieved by using symmetric key encryption and cryptographic hash functions. This technique lacked randomness in the challenges and thus proved to be unreliable as the CSP could cheat by caching the responses. There existed a limitation on the number of times a data owner could query for the integrity of his data. Furthermore, both of these schemes were applicable to only a single copy of data. Curtmola et al. (2008) proposed a scheme, which extended the application of the PDP scheme to multiple replicas of files. This scheme was capable of generating new replicas on demand in case of any corruption or deletion of any replica. It employed masking by utilizing the PRFs (Pseudo Random Functions) and is capable of resisting collusion attacks by storage servers. Application of MR-PDP is limited to a single cloud having multiple storage servers but is not extendable to a multi-cloud environment. All PDP schemes allowed detection of corruption or server misbehavior but lacked the ability to recover lost data after detection. The proof of retrievability (POR) schemes extended the PDP schemes by adding data recovery feature. Shacham and Waters (2008) proposed two POR schemes for static data. 1) BLS Signatures for short query and short response with public verifiability. 2) Pseudorandom functions for longer query and short response, having only private verifiability support. Other characteristics included blockless verification, unforgeability, unbounded queries, however, but these schemes still fail to prevent data leakage. In 2009, Erway et al. (2009) provided complete support for data dynamic operations (insertion, updating, deletion and modification) while

Table 1 – Comparative analysis of data integrity schemes.

Scheme	Characteristics									Implementation	Remarks
	BV	UQ	SD	DR	PP	DDO	PA	MC	Others		
(Nepal et al., 2011)	✓	✓	✓				✓			RSA-based, homomorphic tags	For static data or append only data and not fit for multicloud. Public verifiability is achieved by compromising strong possession guarantee
(Ateniese et al., 2007)	✓		✓			✓				Cryptographic hash function and symmetric key encryption	CSP can cheat by caching responses due to lack of randomness in the challenges. Limited dynamic data operations capabilities
(Curtmola et al., 2008)	✓	✓	✓							Pseudo Random Functions (PRFs) for masking	Not applicable to replication scenario in multicloud environment
(Curtmola et al., 2008)	✓	✓	✓	✓			✓			Scheme 1) BLS Signatures for short query and short response with public verifiability. Scheme 2) Pseudorandom functions for longer query and short response. But only private verifiability supported	Limited to static data and unable to prevent data leakage by verification protocol
(Dodis et al., 2009)	✓	✓	✓			✓				Rank-based authenticated skip lists	Usable for static data only
(Ateniese et al., 2009)	✓	✓	✓	✓					High availability by exploiting redundancy within-server and cross-server	Integrity-protected error-correcting code (IP-ECC) whose construction draws together PRFs, ECCs, and universal hash functions UHF) which serve purpose of MACs	
(Bowers et al., 2009b)	✓	✓	✓	✓		✓	✓			Used RSA, bilinear pairing and Merkle Hash Tree construction for BLS based block tag authentication.	Privacy leakage possible
(Wang et al., 2009)	✓	✓	✓	✓	✓	✓	✓		Efficient handling of multiple auditing tasks simultaneously using bilinear aggregate signatures	Used Homomorphic authenticator and uniquely integrate it with random mask technique to preserve privacy	Overhead in case of failed auditing responses when multiple auditing tasks performed simultaneously
(Zhu et al., 2010)	✓	✓	✓	✓		✓			Constant client storage supporting a fixed file layout. Secure against pollution and replay attacks.	Used same construction of a message authentication code, proposed in (Bowers et al., 2009a) by combination of universal hashing with a PRFs	Increased communication overhead with reduced storage overhead. Not suitable for online storage systems where read operations are extensive.
(Chen et al., 2010)	✓	✓	✓	✓					Robust (able to identify small corruption of data)	Forward error-correcting codes (FECs)	
(continued on next page)											

(continued on next page)

Table 1 – (continued)

Scheme	Characteristics									Implementation	Remarks
	BV	UQ	SD	DR	PP	DDO	PA	MC	Others		
(Shen and Tzeng, 2011)	✓	✓	✓		✓		✓		Less data storage and less bandwidth as compared to POR	Used RSA and Hash functions (SHA1 and SHA-512), Pseudorandom functions and Pseudorandom permutations	
(Lee and Chang, 2011)	✓	✓	✓			✓				Used skip list and hashes	Client needs to store some secret values, thus extra storage needed, e.g. 4 GB file needs 2MB storage
(Luo and Bai, 2011)	✓	✓	✓						Data sovereignty (coupling of stored data authenticity and geographical location)	MAC-based PDP with network delay measurement capabilities	Cannot guarantee that additional copies of data are not instantiated outside of a prescribed geographic area. Does not cater if proxy server is in between client and storage server.
(Popa et al., 2011)	✓	✓	✓	✓		✓			Protects honest service provider against dishonest client	Used authenticated data structure (range-based 2–3 trees) and incremental signature scheme and ECC (error correcting codes)	Low performance because of Error correcting codes
(Zhu et al., 2011)	✓	✓	✓			✓			Robustness (protection against byte or bit level corruption). Broadcast encryption	Reed-Solomon (RS) codes based on Cauchy matrices	
(Barsoum and Hasan, 2012)	✓	✓	✓				✓			Secure pseudorandom numbers (SPRN), Lagrangian Interpolation.	
(Krzywiecki and Kutry, 2012)	✓	✓	✓						Reduced exponentiations to generate proof irrespective of challenge size. Capable of zero knowledge proof and witness hiding (malicious verifier cannot learn about the underlying witness by talking to prover)	Signatures based on Okamoto protocol using pseudorandom permutations and pseudorandom generators.	No support for data dynamic operations. Limited to static data only.
(Sundareswaran et al., 2012)	✓	✓	✓		✓	✓	✓		Support for large no of users without effecting performance of verification process. Traceability (The original user is able to trace a signature on a block and reveal the identity of the signer).	Used group signatures for homomorphic authenticators to provide public auditing. Rely on Homomorphic MACs for storage efficiency. Index hash tables as identifiers of blocks	Unable to identify small corruptions with sampling. Computation cost increases with increase of sampled blocks to identify small no of corrupt blocks
(Yang and Jia, 2012)	✓		✓					✓	Proof of Ownership (Deduplication which means detection and deletion of unwanted duplicate copies). Combined PDP with POW	Merkle Hash Tree, Pseudorandom functions, Hashing	Less efficient and less secure than existing POW schemes but provides the public auditing where existing POW schemes do not.

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keeping the basic properties of verification, unforgeability, and unbounded queries intact. This scheme used a rank-based authenticated skip list to cater dynamic data operations. [Bowers et al. \(2009a\)](#) proposed HAIL, which kept existing features of the PDP schemes intact, and focused on high availability by exploiting the redundancy within a server and across server, along with the provision of data recovery capability. HAIL used integrity-protected error-correcting code (IP-ECC) whose construction draws together PRFs, ECCs, and universal hash functions (UHF), which serve the purpose of MACs. [Wang et al. \(2009\)](#) used RSA, bilinear pairing and Merkle Hash Tree construction for BLS based block tag authentication to achieve public verifiability and data dynamics. However, this scheme lacked privacy protection, which is an essential requirement with public auditability. Among all these schemes, those having the capability of public auditing lacked privacy-preserving feature. [Wang et al. \(2010\)](#) proposed a data integrity scheme that provided public auditing along with privacy protection, by using random masking integrated with homomorphic authenticator. [Wang et al. \(2010\)](#) also introduced the concept of batch auditing, which allows multiple auditing tasks to be performed simultaneously. Nonetheless, it incurred performance overheads when failed auditing responses were received. [Chen et al. \(2010\)](#) used the construction of a message authentication code (MAC), proposed in [Bowers et al. \(2009a\)](#), by combining universal hashing with PRFs. One of the salient features of [Chen et al. \(2010\)](#) was that it incurred less storage overhead though it increased communication overhead. However, it only supported private verifiability and was only applicable for a single cloud. From a security perspective, the scheme in [Chen et al. \(2010\)](#) was secured against reply and pollution attacks. On the contrary, by using network codes, a small portion of a file cannot be read without reconstructing the entire file, thus, network codes based approach is not suitable for online storage systems where the read operation is extensively performed. However, they may be used for backup and archival system. In 2011, [Ateniese et al.](#) improved the capability of PDP schemes by introducing robustness. However, it still lacked support for dynamic operations and public verifiability. The said scheme utilized Forward Error-Correcting Codes (FEC) to achieve data recovery. [Lee and Chang \(2011\)](#) proposed a scheme implemented using RSA-based algorithms and used hash functions (SHA-1 and SHA-512), pseudo-random functions, and pseudo-random permutations. This scheme is efficient in terms of data storage and as well as in its utilization of bandwidth. However, the said scheme is only applicable to static data though it supports public verifiability with privacy protection. In another study, [Liu et al. \(2011\)](#) improved the dynamic PDP performance by using skip lists and hashes. However, this scheme lacked support for public verifiability and suffered from the increase in the client side storage, as it needs to store secret values to perform the hashes. [Peterson et al. \(2011\)](#) extended PDP to include the verification of SLA, used in keeping the data in certain geographical boundaries. The authors have used MACs for the proof of possession and network delay measurement to identify the geo-location of data. However, no guarantee can be provided regarding the instantiation of extra copies outside the prescribed geographic area. The scheme in [Peterson et al. \(2011\)](#) is only applicable to static data, does not support public verifiability, and fails if there is a proxy server between the client

and server. [Zheng and Xu \(2011\)](#) introduced the concept of fairness in data integrity schemes, which means the protection from dishonest client accusing the honest CSP of his data corruption. This scheme used authenticated data structure (range-based 2–3 trees), incremental signature scheme and error correcting codes (ECC), but performance was low due to ECC. This scheme supported dynamic data operations but lacked support for public verifiability. [Chen and Curtmola \(2012a\)](#) achieved robustness using Reed–Solomon codes based on Cauchy matrices to provide bit and byte level data corruption detection. Although the scheme provided robustness for dynamic data operations, however, data recovery and public verifiability are not supported. The scheme proposed by [Krzywiecki and Kutty \(2012\)](#) utilized secure pseudorandom numbers (SPRN) with Lagrangian interpolation to improve computation and storage efficiency. Hence, it can be compatible with smartphones and embedded systems with limited resources. In another study, [Mohan and Katti \(2012\)](#) achieved reduced exponentiations to generate proof of possession irrespective of challenge size. The scheme is based on zero knowledge proof and witness hiding (malicious verifier cannot learn about the underlying witness by talking to prover). The signature scheme used is based on Okamoto protocol using pseudorandom permutations and pseudorandom generators. However, no support for data dynamic operations and public verifiability is provided. The scheme proposed by [Wang et al. \(2012\)](#) supports public verifiability of remote outsourced storage shared by multiple users in a privacy-preserving manner (privacy protection of user's data is provided against external auditors). In order to support public auditing, group signatures are used for homomorphic authenticators and homomorphic MACs for storage efficiency. An index hash table holds identifiers of blocks. Despite providing privacy preservation, the scheme is unable to identify small corruptions with sampling and computation cost increases with the increase of sampled blocks. Traceability is supported for large number of users without affecting the performance of the verification process. Here, traceability means that the original user can trace a signature on a block and reveal the identity of the signer. [Zheng and Xu \(2012\)](#) utilized Merkle Hash Tree (MHT), pseudorandom functions and hashing. Provable data possession (PDP) and proof of ownership (POW) are combined along with deduplication, which means detection and deletion of unwanted duplicate copies. Furthermore, the scheme supported public auditing, which was missing in previous schemes. [Zhu et al. \(2012\)](#) introduced high security, high performance, and transparent verification for PDP model in the context of multi-cloud. This scheme is based on homomorphic verifiable response (HVR) and hash index hierarchy (HIH), and presented a cooperative PDP (CPDP) scheme using bilinear pairings. It supports all characteristics of data integrity schemes such as blockless verification, unforgeability, unbounded queries, data recovery, public auditing, privacy protection and data dynamic operations. However, [Zhu et al. \(2012\)](#) failed to provide knowledge soundness and is unable to resist against data leakage, and tag forgery attacks as pointed by [Wang and Zhang \(2014\)](#). [Shi et al. \(2013\)](#) reduced bandwidth overhead by using a Merkle Hash Tree (MHT). Other major contributions of this work are independent metadata size, reduce disk seeks, and support for dynamic data. [Sarkar and Safavi-Naini \(2013\)](#) utilized raptor

codes (a special class of fountain codes) and supported data recovery in case of corruption. The uniqueness of this scheme is that the numbers of blocks in a challenge are not fixed. [Gondree and Peterson \(2013\)](#) combined the data integrity verification with constraint-based data geolocation using latency-based techniques in multi-cloud environment. However, this scheme is not suitable in a setting where a private network of significantly better quality than the Internet connects remote sites. [Esiner et al. \(2013\)](#) proposed an approach that was the first to use variable length for input file data blocks. It employed authenticated skip list based FlexList and RSA-based homomorphic tags with support for data dynamic operations and multiple audit requests handling simultaneously. [Chen \(2013\)](#) utilized algebraic signatures to take advantage of low network bandwidth, with reasonable computation load. The scheme provided resistance to malicious modification, although it still suffered from limitations such as signatures used not being cryptographically secure, no support for data recovery, public verifiability and the lack of support for dynamic data operations. [Cash et al. \(2013\)](#) combined the concept of oblivious RAM with erasure codes to hide the access pattern hence avoiding correlation of challenged blocks to physical storage. This scheme can efficiently update redundantly encoded data. However, it introduced high network communication costs. [Wei et al. \(2014\)](#) focused on secure computation and used commitment-based sampling (CBS) using Merkle hash tree, bilinear pairings, and designated verifier signatures. This scheme lacks dynamic data operations and data recovery capability but supports public verifiability. [Zhu et al. \(2012\)](#) presented an efficient PDP scheme where multiple CSPs cooperatively store and maintain the clients' data. Then, based on homomorphic verifiable response (HVR) and hash index hierarchy (HIH), they presented a cooperative PDP (CPDP) scheme from the bilinear pairings. They claimed that their scheme satisfied the security property of knowledge soundness. However, [Wang and Zhang \(2014\)](#) performed cryptanalysis of this scheme and argued that it lacked knowledge soundness.

We perform a comparative analysis of all the schemes above in [Table 1](#). The analysis reveals that all data integrity schemes have three major concerns, which are (a) computation, (b) communication, and (c) storage efficiency. There is always a tradeoff between computation, communication, and storage efficiency. Besides the said tradeoffs, security of data integrity schemes becomes a challenging problem when dynamic data operations (insert, update, and delete) are supported. Public auditing and access control in a multiuser environment is also another important issue handled by data integrity schemes. The PDP schemes have focused on verification. Identification of bit or byte level corruptions using PDP schemes is a challenging task due to probabilistic nature of these schemes. To identify bit or byte level corruptions with 100% guarantee, we have to verify the complete file, whereas verification is performed through random sampling in the case of a probabilistic approach. On the other hand, the POR schemes support verification and data recovery. Selection of metadata primitives is a major issue of data integrity schemes. Metadata primitives, chosen for implementation of both PDP and POR schemes significantly affect storage and computation efficiencies. A cryptographic primitive provides security but incurs increased computation overhead, whereas other non-cryptographic ([Chen,](#)

[2013](#)) primitives are computation efficient but lack security. Details of design challenges of data integrity schemes are discussed in [Section 4](#). From the deployment environment perspective, said schemes are mostly applicable for a single cloud environment, whereby only a few have focused on multi/hybrid cloud environment. Most of the existing schemes can handle a single copy of data, and very few are applicable in replication scenarios. Therefore, one future aspect of data integrity schemes would be handling of multiple replicas of data and issues associated with it (such as proof of ownership, deduplication). We have discussed such future trends of data integrity schemes in [Section 5](#). See [Table 1](#) below for the comparative analysis of data integrity schemes.

5. Design challenges of data integrity schemes

Main design challenges of data integrity schemes are discussed below:

5.1. Computation efficiency

All data integrity schemes preprocess the data before outsourcing it to a cloud storage server, similarly, at the cloud storage server; metadata are generated from original data. The overhead of performing such processing may affect the computation efficiency of data integrity schemes. Computation efficiency of preprocessing phase does not matter for small datasets, but it has a serious impact for large datasets. Computation cost on the server side for the proof generation limits on how frequently user can verify the integrity of outsourced data. Primitives used by the data integrity scheme as metadata also impact computation time. For example, algebraic signatures ([Chen, 2013](#)) based metadata provide an improved computation efficiency of verification process as compared to cryptographic primitives. In the case of POR schemes, network codes ([Chen et al., 2010](#)) provide better efficiency than the error correcting codes. [Han et al. \(2014\)](#) have used maximum rank distance (MRD) codes ([Han et al., 2014](#)) to generate homomorphic tags, which improved the server-side computation. [Chen et al. \(2014\)](#) have utilized Hill cipher for integrity verification with fast computation. [Kwon et al. \(2014\)](#) have reduced the auditing cost with the application of new index table management.

5.2. Storage efficiency

For integrity verification and data recovery by any data integrity scheme, additional metadata is required along with the original data. The above storage overhead on client and server side describes storage efficiency. In [Chen et al. \(2010\)](#), the client needs to store some secret values thus requiring an extra storage, such as for a 4 GB file, a 2MB storage is needed. [Chen et al. \(2014\)](#) managed to produce small size metadata and improved the storage cost. Furthermore, data de-duplication ([González-Manzano and Orfila, 2015](#); [Latesh Kumar and Lawrance, 2015](#); [Miao et al., 2015](#)) plays a significant role in the reduction of storage though will incur some cost.

5.3. Communication efficiency

Three aspects describing the communication efficiency of data integrity schemes are (Wang, 2014): (1) Overhead of the initial data transfer along with metadata, (2) challenge requests for possession verification by data owner and (3) challenge response for proof of possession sent back by the cloud storage server. In the case of dynamic data, this communication overhead also includes the updates verification. Primitives used for metadata have an impact on the communication cost. Algebraic signatures (Chen, 2013) provide communication efficiency through low network bandwidth utilization during challenge request and response time. By using Hill cipher (Chen et al., 2014), the size of challenge and response is usually small, providing that communication is efficient. In the case of multiple replicas, MuR-DPA (Liu et al., 2014) scheme improved communication for both update verification and integrity verification.

5.4. Reduced disk I/O

Disk I/O efficiency of a data integrity scheme is derived by the overhead in block access and metadata access for proof generation on cloud storage server. Accessing all blocks for proof generation impacts the efficiency of a scheme and makes the scheme impractical for large datasets.

The following parameters influence the overall efficiency of any disk I/O:

- **Size of block:** Data integrity schemes either use fixed (Ateniese et al., 2007) or variable length block size (Esiner et al., 2013). The smaller the size of the block, the larger the number of blocks in a file. This will impact preprocessing time in generating metadata tags for all blocks. Depending on this parameter, the efficiency of a scheme is significantly affected by the variable length of metadata, the number of blocks in the challenge and unordered indexes in the challenge request.
- **Variable length of data/metadata:** The data or metadata for a particular block index cannot be directly accessed, due to variable length. The searching of data or metadata in the above case is similar to searching in a link list. To reach a particular index data or metadata, preprocessed file has to be traversed sequentially, skipping undesired blocks/tags. As a result, this increases disk I/O as the number of block tags increases in the verification process, hence proof generation time increases (Esiner et al., 2013). Schemes have improved this searching using a tree-like data structures, such as authenticated data structure (skip list), balanced tree, and hash indexes. The performance of a scheme can be significantly improved by reducing the sequential access to direct access. By reducing the sequential access to direct access, a significant performance gain can be achieved.
- **Number of blocks in challenge:** This parameter has an impact on both computations cost and I/O cost. A large number of blocks in a challenge will lead to increased time in generating the proof (Sarkar and Safavi-Naini, 2013). Implementation primitives used as metadata directly affect computation cost. An important property of unbounded queries (no limit on how frequently user can verify its data) is lost for large datasets depending on this parameter.

Okamoto protocol based signatures (Mohan and Katti, 2012) have been used to reduce exponentiations needed in generating proofs, independent of the number of blocks in challenge thus improving computational efficiency.

- **Unordered block indexes in challenge:** If indexes of blocks are in ascending order in the challenge generated by a client, then verification time can be improved (Shi et al., 2013). The reason is that searching for each metadata tag in a challenge will not require the traversal of the metadata file from the start; hence, the server can continue to look for the next index in the challenge from the current position. With this parameter, disk I/O is significantly reduced, improving proof generation time.

5.5. Security

Data integrity schemes are vulnerable to a variety of attacks. Therefore, care must be taken in designing such a scheme. Following are the possible attacks that may be launched against a data integrity scheme along with their possible solutions:

- **Tag forgery attack:** Forgery of tags is possible by malicious CSP in an attempt to hide the user's data damage and bypass the auditing challenge. A CSP may try to forge the tags to deceive the verifier successfully (Yang and Jia, 2012; Yu et al., 2014; Zhu et al., 2012). Cryptographic homomorphic tags can provide unforgeability (Ateniese et al., 2007; Zhu et al., 2011).
- **Data deletion attack:** CSP may pass the challenge by generating a valid proof of possession using tags only where original data might have been deleted entirely (Yu et al., 2014).
- **Replace attack:** CSP may replace the corrupted or deleted pair of data blocks and respective tags with another valid pair in the response of a challenge, whereby deceiving the verifier (Yang and Jia, 2012; Yu et al., 2014).
- **Replay attack:** The CSP may cache the previous challenge responses of verification phase and eventually replays the cached responses for new challenges matching it. In the context of remote storage, the term such as "rollback attack" or "forking attack" is also used for indicating replay attacks (Tate et al., 2013).
- **Pollution attack:** In pollution attack, a dishonest server uses correct data in response generation against a challenge but provides corrupted blocks in repair phase for coding new blocks (Chen et al., 2010).
- **Data leakage attack:** In this type of attack, an adversary can extract stored data during the verification protocol using techniques such as wiretapping (Zhu et al., 2011, 2012).

In the presence of such attacks, data integrity schemes may fail in timely identification of data corruption and result in an unrecoverable damage. Following guidelines may be used in designing a data integrity scheme, which can resist against the described attacks:

1. Metadata should be tamper evident (Ateniese et al., 2007).
2. Proof generation mechanism must utilize the data blocks along with metadata (Ateniese et al., 2007).
3. Proof of possession should be semantically secure (Zhu et al., 2011).

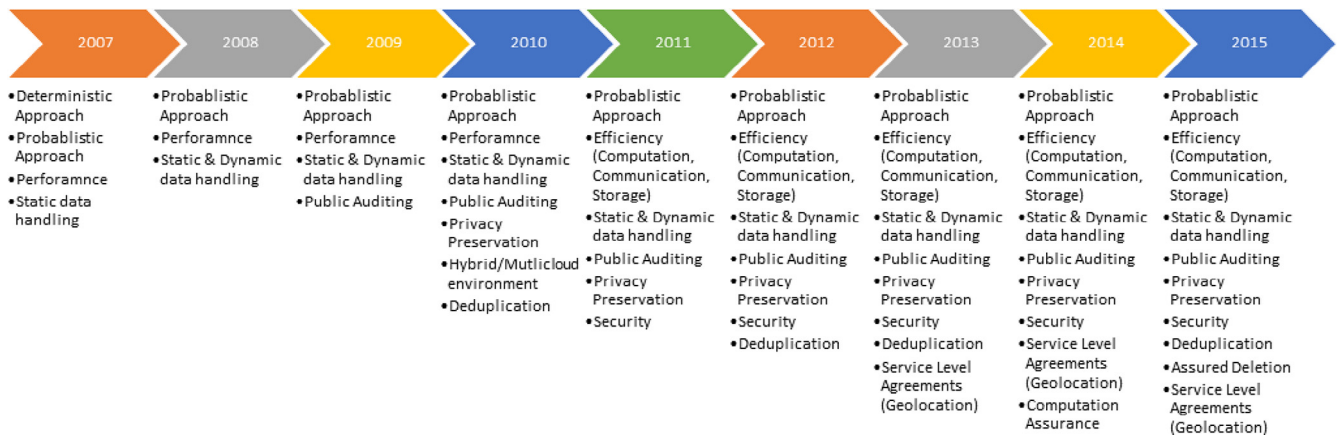


Fig. 4 – Data integrity schemes evolution timeline.

- Metadata and its corresponding data block must have strong binding with each other (Yang and Jia, 2012; Yu et al., 2014).
- Randomness must be used in the challenge–response protocol (Ateniese et al., 2007).

These simple characteristics can make a data integrity scheme more secure and reliable. Verification protocol of data integrity schemes rely on the security of metadata.

If metadata are not secured, then this raises concerns over the security and reliability of the underlying scheme. To hide the data corruptions, malicious CSP may modify metadata to pass the challenge phase. If a malicious CSP can do that successfully without being detected, then the underlying scheme is not secured and reliable. This case justifies the requirement of having a tamper evident metadata (Ateniese et al., 2007) to identify a malicious CSP. Recommendations (2) and (3) are related and collectively define the strength of the data integrity scheme. If a scheme solely relies on the metadata for generation of proof of possession, then CSP may delete the data and keep only the metadata. So utilizing data and metadata in the proof generation can mitigate the data deletion attack. However, if proof of possession is not semantically secured, then challenge–response phase is vulnerable to data leakage attack. Therefore, while designing a data integrity scheme, care must be taken so that the metadata used for verification may not result in leakage of original data during the verification protocol. Modern cryptographic techniques such as “Multi-Prover Zero-Knowledge Proof System” (Zhu et al., 2011) can provide protection against tag forgery attack and data leakage attack. Using homomorphic verifiable responses and hash index hierarchy “Collaborative PDP” (Zhu et al., 2011) ensures that any adversary capable of eavesdropping the communication channel will not be able to extract any information from data exchanged in verification protocol. Metadata must be tamper-evident and proof generation process must utilize data and the corresponding metadata; however, if data and metadata have no strong binding then still a scheme may be vulnerable to replace attack. By establishing a strong binding between data blocks and corresponding metadata, replace attack can be mitigated. Furthermore, replace attacks can also be avoided using authentication of block indexes (Yu et al., 2014). The last recommendation is randomness in the challenge–response

protocol. In the absence of randomness, a scheme may be prone to replay attack, i.e. CSP replaying cached responses of past challenges (Chen et al., 2010; Erway et al., 2009; Yang and Jia, 2012) in response to new challenges.

6. Discussion

We have identified the trend of evolution of data integrity schemes by surveying existing schemes. Fig. 4 depicts the timeline, which presents the scope of data integrity schemes that keep on extending with the passage of time. Research on data integrity schemes started before 2007. However, Ateniese et al. (2007) proposed first the formal method for provable data possession in 2007. Due to the deterministic nature of the scheme, performance was a major issue at that time. PDP (Ateniese et al., 2007) was the first probabilistic approach that improved performance by random sampling of blocks in the verification process. Up to 2007, all schemes handled static data only. In 2008, the focus was shifted to dynamic nature, where (insert, update, delete operations) were supported (Ateniese et al., 2008; Chen et al., 2010; Curtmola et al., 2008; Erway et al., 2009; Wang et al., 2009, 2010; Zhu et al., 2010). While handling dynamic data, different attacks (e.g. replay attacks; Erway et al., 2009) were identified in the schemes proposed at that time. Attacks on integrity schemes lead the research to shift on the security of data integrity schemes themselves. With the usage of authenticated data structures such as skip list and Merkle hash tree, data integrity schemes were able to handle dynamic data securely and efficiently (Esiner et al., 2013). For large datasets, the performance of data integrity schemes was still a major issue. However, new aspects kept on evolving in the context of data integrity schemes. In Ateniese et al. (2007), the concept of public verifiability was discussed with weaker possession guarantees. Later in 2009 (Wang et al., 2009), secured public verifiability became a part of data integrity. With the introduction of public verifiability, privacy issues were identified, and in 2010 privacy-preserving public verifiability (Wang et al., 2010) became an essential part of data integrity. Before 2010, all research on data integrity schemes had focused on a single cloud environment. In 2010, for the first time, data integrity was discussed in the context of hybrid or multi-cloud (Zhu et al.,

2010). During 2010–2012, major focus areas of data integrity schemes were efficiency (computation, communication, and storage), dynamic data, and security. In 2011, service level agreement (Popa et al., 2011) verification emerged as a new feature of integrity checking. Geolocation of data (Peterson et al., 2011) was also included as an aspect of service level agreement verification. Later on, assured deletion (Habib et al., 2013) and re-outsourcing of data (Jiang et al., 2014) were also discussed. In 2014, computation assurance (Wei et al., 2014) aspect was highlighted in the context of secure data integrity. Since 2015, the deduplication (Bellare et al., 2013a, 2013b; González-Manzano and Orfila, 2015; Latesh Kumar and Lawrance, 2015; Miao et al., 2015; Zheng and Xu, 2012) issues are considered necessary for large datasets, as storage cost and access efficiency are significantly affected in replication scenarios.

7. Future trends of data integrity schemes

Cloud computing model is evolving dynamically, and different business and cloud usage aspects are emerging. New trends have broadened the scope of data integrity schemes. The following are the recently emerging research trends in the context of data integrity schemes.

1. **Geo-location assurance:** Service level agreements may restrict the CSP to keep data, in a particular geographic region at the granularity level of city, state, time zone, or political boundary. For compliance purpose, some cloud users are required to ensure geolocation of data (Albeshri et al., 2014; Fu et al., 2014; Gondree and Peterson, 2013; Hildebrandt, 2014; Paladi et al., 2014; Peterson et al., 2011). Cloud users cannot trust the CSP. A careless or naïve CSP may migrate user data to an overseas data center violating the SLA in an attempt to leverage cheaper IT cost. Such actions of the CSP can make user data available for foreign governments indirectly who can inspect it through search warrants or any legal mechanism. In this scenario, timely identification of intentional dishonesty or violation of SLA by the CSP is an essential requirement for the data owner. New research on data integrity schemes is incorporating geolocation verification.
2. **Assured deletion:** In cloud storage, data deletion should be guaranteed otherwise it could result in breaches (Tang et al., 2012). Another important aspect in assured deletion is that other versions, which share the common data of the deleted versions or files, should remain unaffected. Rahumed et al. (2011) proposed “FADE” which is a version controlled backup system with the capability of fine-grained assured deletion. The deleted versions are ensured to be permanently inaccessible. SoK (Reardon et al., 2013) highlights the approaches for secured, assured deletion and taxonomy of adversaries and their capabilities to take advantage of cloud storage in the absence of a secured deletion. Assured deletion is also termed as self-destruction (Cachin et al., 2013; Chaoling et al., 2014; Habib et al., 2013; Xiong et al., 2013).
3. **Deduplication:** Deduplication is the detection and deletion of unwanted duplicate copies to improve storage cost. Deduplication can be at the file level or block level. Side channel attacks (Harnik et al., 2010) have been launched through the deduplication process. Benefits of deduplication include reduced backup cost, lower cost of business continuity or disaster recovery, increased storage efficiency and enhanced network efficiency. Simplest approach used for deduplication is to compare objects (file or blocks of a file) and remove copies, which already exist in storage (Bellare et al., 2013a, 2013b; González-Manzano and Orfila, 2015; Latesh Kumar and Lawrance, 2015; Miao et al., 2015; Zheng and Xu, 2012). For identification of copies, hashes of files or blocks are computed and stored, and if hash for any new block is found to be present in the stored hashes, then the copy is removed.
4. **Computation assurance:** Computational reliability is a future requirement for any efficient CSP. Computation assurance can increase the level of confidence of cloud users for outsourcing their data storage. Google cloud platform has annual audits from standards like ISO 27001 and FISMA to guarantee the assurance. But the user cannot solely rely on the guarantees claimed by CSP. Some attempts have been made on data integrity schemes to ensure computation reliability. These include approaches with a trusted hardware component (Tate et al., 2013) such as “Trusted Platform Module (TPM)” and without a hardware component (Wei et al., 2014). Whereas, Khan and Hamlen (2013) allow untrusted, mostly serial computations in untrusted computing environments to be independently and efficiently validated by trusted, commodity clouds.
5. **Self-controlling objects (SCO):** Some attempts have been made to encapsulate sensitive resources along with their access control policies in programmable Java Achieves (JARs). This approach assured the protection of sensitive resources through the provision of adaptive security policies (Squicciarini et al., 2013). This idea of SCO can be extended to achieve geolocation, assured deletion, and deduplication in the future.
6. **Re-outsourcing data:** Data integrity schemes are incorporating verification of SLA, and they are a practical requirement of cloud users, which CSP must satisfy and ensure. Client data should not be re-outsourced without prior knowledge and consent of client (Jiang et al., 2014). Re-outsourcing detection is indirectly a necessary part of data integrity when outsourcing data to a multi-cloud environment.

8. Conclusion

Cloud storage is a cost effective solution and provides elasticity, pay-as-you-go, and business continuity with long-term retention and risk mitigation through disaster recovery. These features are not available with on-premises storage. However, cloud storage model has inherent weaknesses; a user outsourcing data loses control over his data. Data confidentiality and integrity are the major concerns of the user when moving data from on-premises storage to the cloud storage. Many data integrity schemes have been proposed over the years to address the security issues associated with cloud storage. In this paper, we have discussed cloud computing and cloud storage. We have also discussed the cloud storage issues. Our focus in this paper was to improve the understanding of security aspects of data integrity schemes. Schemes proposed over the years are not suitable for all types of data and environments. Therefore, we

establish taxonomy based on attributes that can help the readers to understand different aspects (nature of data, nature of deployment environment, etc.) of data integrity schemes. The said attributes impact the applicability and efficiency of schemes in practical scenarios. We provide a comparative analysis based on characteristics of data integrity schemes. We establish the timeline of the evolution of data integrity schemes and also identify future aspects that researchers can exploit to perform their future research. We identify the design challenges in performance and security of data integrity schemes. We also identify performance-degrading factors of data integrity schemes and provide a discussion on mitigation strategies for the possible attacks.

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